

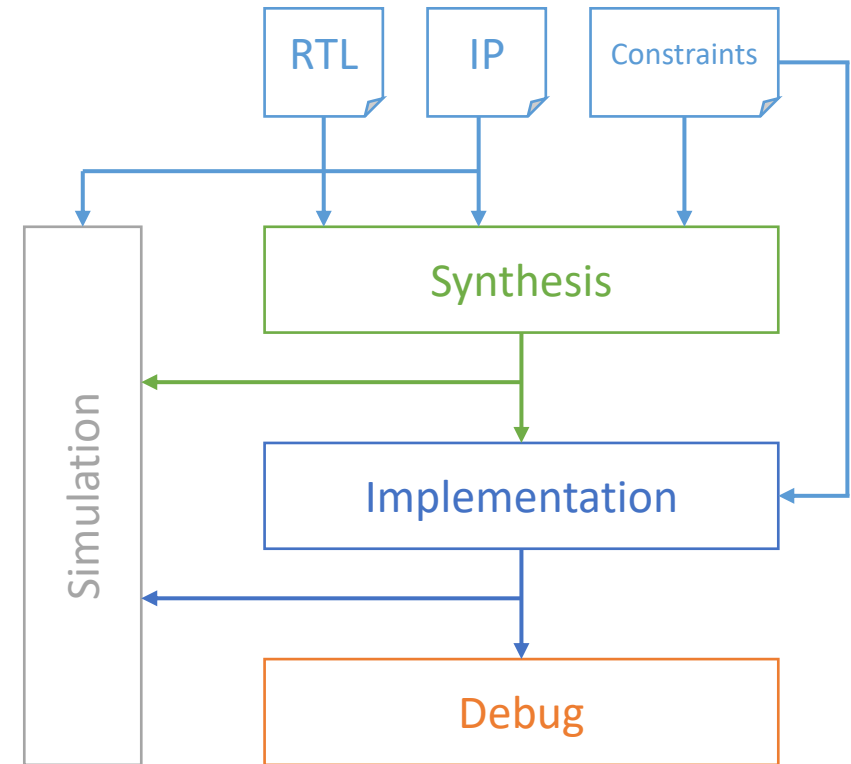
FPGA Programming – VHDL Part

VHDL by Examples – Debugger

Andrea Triossi – University of Padova – INFN Padova

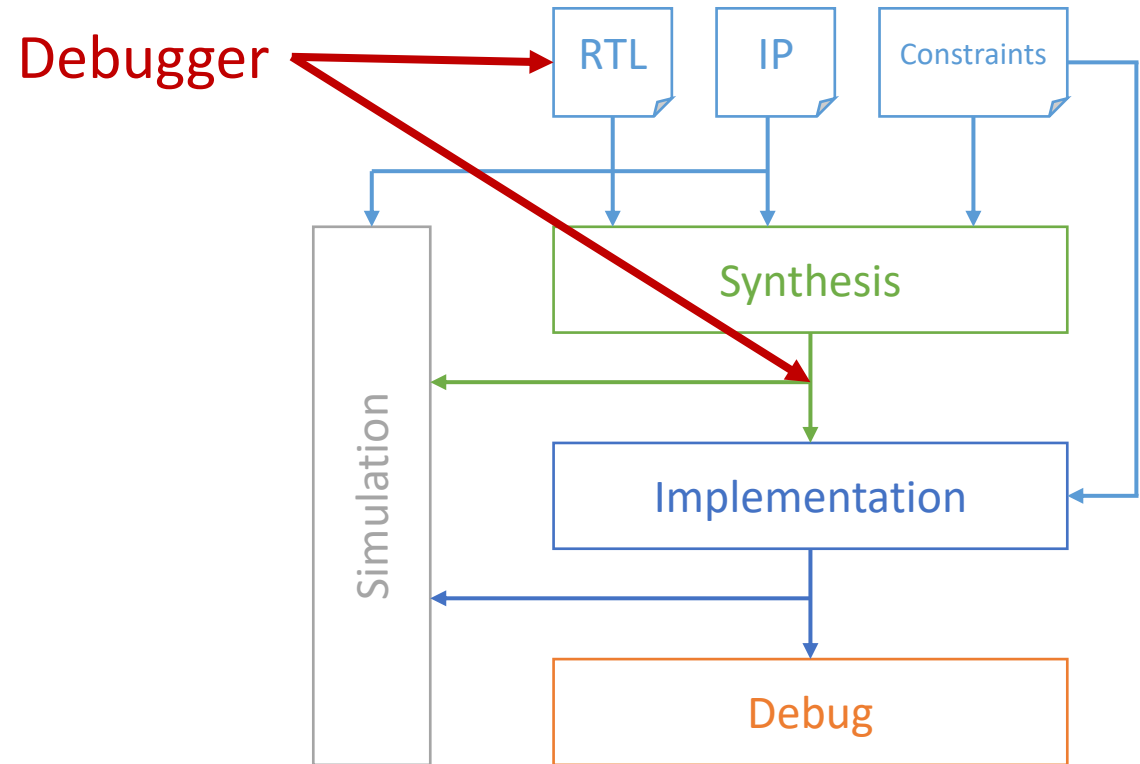
Design Flow

- RTL source files
- Intellectual Property (IP)
- Constraint files
- Simulation
- Synthesis
- Implementation
- Debug



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Debugger

Can be inserted in two ways:

- In the RTL source files
 - + The names of the scoped nets are preserved
 - Your circuit is “altered” by the debugger
- In the netlist (after synthesis)
 - + Your circuit is synthesized independently
 - The name of the nets to be spayed is assigned by the synthesizer

Debugger in RTL

- Generate an IP with the help of the IP wizard
 - ILA (Integrated Logic Analyzer)
 - Monitor the internal signals
 - Customizable data width depth
 - Customizable trigger rules: boolean or sequential conditions (TSM language)
 - VIO (Virtual Input/Output)
 - Monitor or drive internal signals
 - Virtual led and virtual switch/button
- Instantiate the generated IP in one of your RTL files

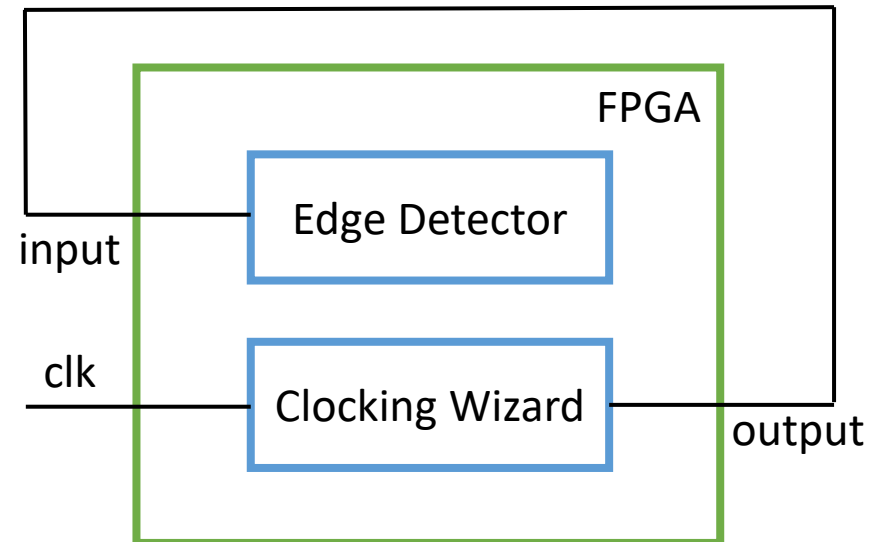
Debugger after synthesis

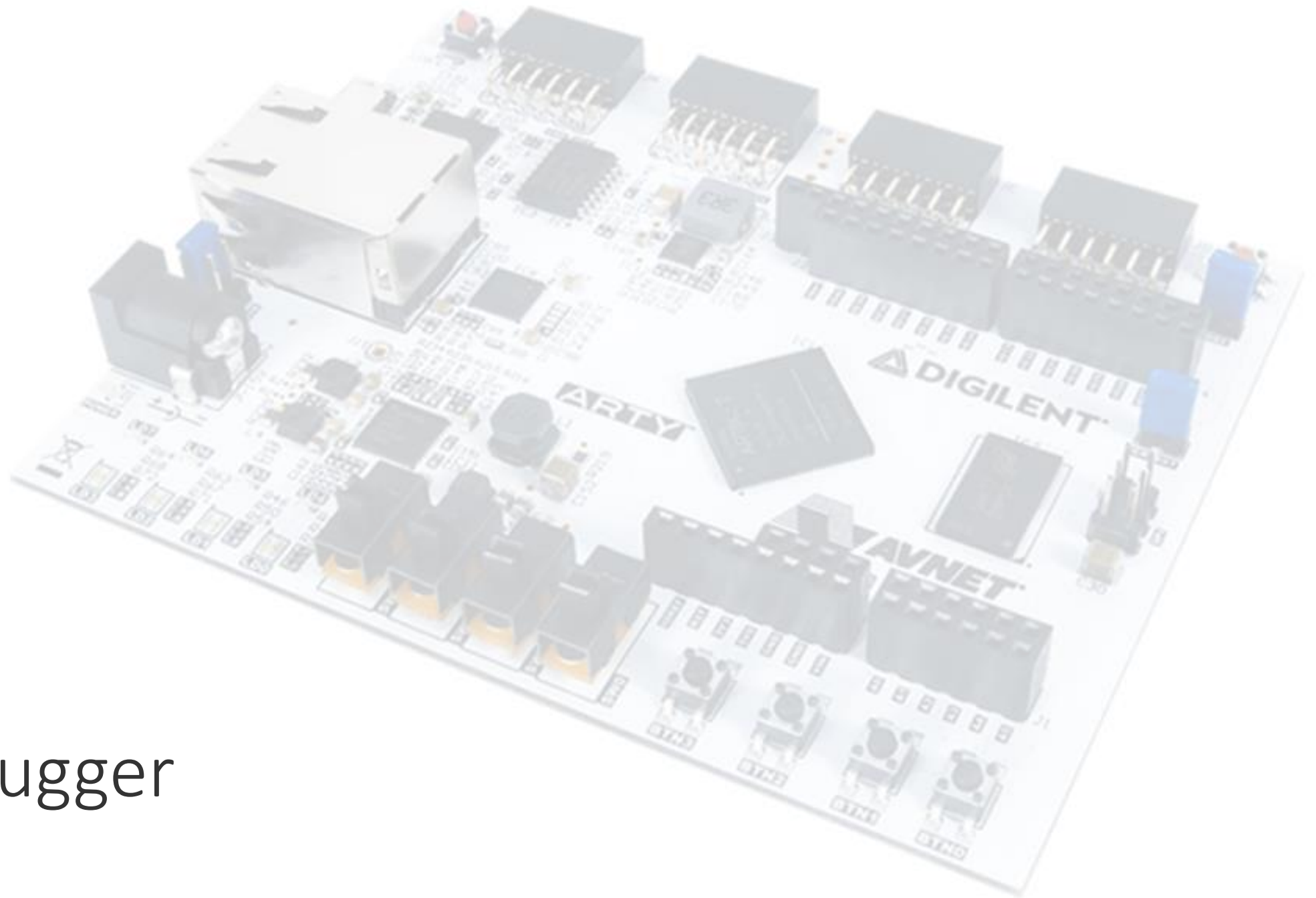
- Synthesize your design alone
- Set-up debug probes in the netlist
- Generate a new constraint file
- Resynthesize and implement
- This works only for ILA
- Whenever you want to remove ILA, you can disable the generated constraint file

Exercise

- Design an edge detector for an input signal
- Generate a **non-integer-related** clock using the Clocking Wizard IP
- Route the generated clock to an output pin, then loop it back using an external wire
- Add debugger to monitor input, internal and edge
- Capture and verify the debug traces
- Improve robustness by adding a double register synchronizer to avoid [metastability](#)

```
process(clk)
begin
    if rising_edge(clk) then
        internal <= input;
        if (internal = '0') and (input = '1') then
            edge <= '1';
        else
            edge <= '0';
        end if;
    end if;
end process;
```





EX Debugger